PUSHKAR KUMAR

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EDUCATION

Indian Institute of Technology Roorkee, India *2023 – 2028*
BS-MS Physics

EXPERIENCE

Research Collaboration | Remote Independent Research | repo *July 2026 – Present*
Mentor: Dr. Deepak Kumar, Postdoc at KDDI Research Inc., Japan

- Exploring **test-time adaptation** for **speech and multimodal emotion recognition**, focusing on robustness under speaker, domain and modality shifts.
- Benchmarking **3B-scale VLMs** on multimodal emotion datasets by extracting joint video-text embeddings and training lightweight **MLP classifiers** for emotion recognition.

Research Intern | QIntern 2026 (QWorld) *July 2026 – August 2026*
Mentor: Abhishek Sawaika, PhD at University of Melbourne, Australia

- Developed a **Java based quantum networking module** for the **iQuantum simulation toolkit** to model distributed quantum computing across connected quantum processors.
- Modeled **entanglement generation**, **resource reservation**, routing and node-to-node communication for quantum-networked execution.
- Benchmarked a **two-node quantum network** across distance, attenuation, detector efficiency and memory efficiency to evaluate entanglement generation throughput.

Summer Research Intern | Physics Department, IIT Roorkee | repo *May 2026 – July 2026*
Supervisor: Prof. Mayank Goswami

- Developed a **deep learning based CT reconstruction pipeline** for recovering 3D volumes from sparse and noisy projection data.
- Implemented a **dual-domain neural architecture** combining sinogram-domain denoising, physics-inspired reconstruction priors and image-domain **UNet** refinement.
- Simulated CT projection datasets using **gVXR ray tracing** with realistic imaging physics, noise models and density variations for supervised reconstruction experiments.
- Integrated **PyTorch** training workflows with edge-aware losses, reconstruction-quality evaluation and a **PyQt5 GUI** for model training and inference.

OPEN SOURCE CONTRIBUTIONS

LLVM Compiler Infrastructure Repository | Pull Requests

PROJECTS

RISC-V ISA Simulator | *C++* | *Computer Architecture* | github *July 2026 – September 2026*

- Built a modular **RV32I ISA simulator** in **C++17** modeling instruction loading, fetch, decode, execution, memory access, register write-back and program-counter updates using a **single-cycle** CPU model.
- Added support for loading hexadecimal machine-code programs, executing arithmetic, logical, memory, branch and jump instructions, and printing clear stage-wise traces for debugging CPU behavior.

UniCloak | *DeFi* | *ZK* | *Cross-Chain* | github *Dec 2025 – Jan 2026*

- Built a **zero-knowledge privacy layer** for stablecoin transactions, private swaps, bridging, and cross-chain swaps from a **unified liquidity pool** on **Mantle mainnet**; designed **Noir circuits** for zk-ID, wallet commitments, deposits, and withdrawals.
- Implemented **Solidity** contracts for an identity mixer, private pool, verifier contracts, and a Base **CrossMessenger**; integrated **WooFi**, **Stargate**, a **Node.js relayer**, and an off-chain **Merkle proof** manager to execute private DeFi actions without exposing user wallet identity.

C2F.Pay | *Blockchain* | *AI* | github *August 2025*

- Built an **AI-orchestrated crypto-to-fiat checkout** where **Claude Desktop** coordinates Merchant, Relayer, and Sei Wallet **MCP** servers to discover products, create **Stripe PaymentIntents** and complete purchases from natural-language prompts.
- Implemented an **escrow flow** using a **Locker smart contract**, on-chain Locked events, Stripe payment verification, resolver settlement and receipt notifications so user funds remain escrowed until fiat payment finality is proven.

Stoxly | *Blockchain* | *AI* | devfolio *May – June 2025*

- Built a **tokenized U.S. stocks** protocol on **Avalanche Fuji**; used **Chainlink Functions** and **Chainlink Price Feeds** to fetch stock prices, convert AVAX/USD values, and execute mint/redeem flows backed by off-chain **Alpaca API** orders.
- Added **Chainlink CCIP** support for cross-chain USDC mint/redeem requests from Sepolia to Avalanche Fuji, and connected an **ElizaOS** agent that checks balances and initiates stock purchase or redemption transactions through natural-language wallet commands.

AWARDS / SCHOLARSHIPS / ACADEMIC ACHIEVEMENTS

- **First Prize** worth **$10,500** in the **ElizaOS** track at the **Chainlink Chromion Hackathon**.
- **Second Prize** worth **Rs. 75,000** (approx. **$790**) in the final round of the **TriHacker Hackathon** by The Residency – Bangalore House.
- **Second Prize** worth **Rs. 25,000** (approx. **$260**) in the first round of the **TriHacker Hackathon** by The Residency – Bangalore House.
- **AgentKit Pool Prize** winner at the **Agentic Ethereum** hackathon.
- Selected in the **Top 30 Projects** for Demo Day at the **Mantle Global Hackathon**.
- Selected as one of **38 developers** out of 581 applicants in **PLDG Cohort-4** to contribute to the **Protocol Labs** ecosystem.
- **All India Rank 8,950** in **JEE Advanced 2023**.
- **98.57 Percentile** in **JEE Main 2023**.

POSITIONS OF RESPONSIBILITY & ACADEMIC SERVICE

Reviewer | NeurIPS Sim2Science 2026 Workshop *September 2026*

- Reviewed research submissions involving **simulation-driven scientific machine learning**.

Core Member | Blockchain Society, IIT Roorkee | website | github *August 2025 – Present*

- Student-led **blockchain** community at **IIT Roorkee** focused on **Web3** research, developer collaboration, workshops, and technical meetups.